

Bayesian Analysis of NBA 2023-24 Player Performance: Hierarchical Models for 3-Point Accuracy and Scoring

George Aidinis STAT 9270 - Spring 2025

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Abstract

We analyse 314 NBA 2023-24 player-seasons with two hierarchical Bayesian models. (i) A position-level random-intercept *Binomial-logit* model infers each player’s true 3-point percentage; (ii) a heavy-tailed Student-*t* random-slope model assesses the effect of age on points per game (PPG). We report posterior shrinkage, position differences, predictive checks, and information criteria. All computation was done in R (**brms**); data, code and figures are reproducible.

1 Data

The 2023-24 season totals table was scraped from basketball-reference.com (9 May 2025). Players with <20 games or <30 3-point attempts were removed; for multi-team seasons only the league-aggregate “TOT” line was kept, leaving $N = 314$ rows. Key variables are listed in Table 1; sample sizes by position appear in Table 2. Figure 1 shows the empirical distributions of raw 3-pt percentage and PPG.

Variable	Description
<code>threes_made</code> , <code>threes_trials</code>	Counts of made and attempted three-point shots (likelihood for Model 1)
<code>ppg</code>	Points per game (PTS/ <i>G</i>), response for Model 2
<code>pos</code>	Roster position factor: C, PF, SF, SG, PG
<code>age_sc</code>	Standardised age: $(\text{Age} - \bar{\text{Age}})/s_{\text{Age}}$

Table 1: Variables retained after cleaning

pos	N players
C	31
PF	67
SF	77
SG	81
PG	58

Table 2: Filtered sample sizes by position

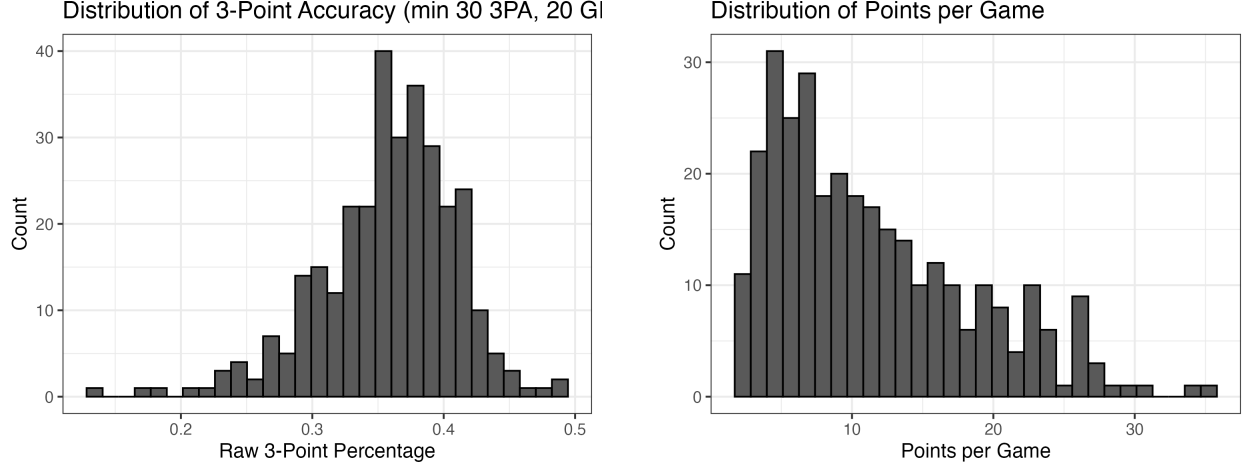


Figure 1: Empirical distributions of 3-pt percentage (left) and PPG (right)

2 Models

2.1 Model 1 - Hierarchical Binomial for 3-pt accuracy

For player i ($i = 1, \dots, N$) in position p_i :

$$y_i \mid \theta_i \sim \text{Binomial}(n_i, \theta_i), \quad \text{logit } \theta_i = \alpha_{p_i} + \beta \text{ age_sc}_i.$$

Position intercepts follow $\alpha_p \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2)$ with weakly informative hyperpriors $\mu_\alpha \sim \mathcal{N}(0, 5)$, $\sigma_\alpha \sim \text{Half-Normal}(0, 1)$, and $\beta \sim \mathcal{N}(0, 2)$.

2.2 Model 2 - Student- t random-slope for PPG

$$y_i \sim t_\nu(\mu_i, \sigma), \quad \mu_i = \alpha_{p_i} + \beta_{p_i} \text{ age_sc}_i, \quad (\alpha_p, \beta_p)^\top \sim \mathcal{N}_2(\mathbf{0}, \Sigma).$$

$$\underbrace{\alpha_p \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2), \beta_p \sim \mathcal{N}(\mu_\beta, \sigma_\beta^2)}_{\text{second level}}, \quad (\mu_\alpha, \mu_\beta) \sim \mathcal{N}(\mathbf{0}, 10^2 I), \quad (\sigma_\alpha, \sigma_\beta) \sim \text{Half-N}(0, 2).$$

Priors for the likelihood scale and degrees of freedom are $\sigma \sim \text{Half-Normal}(0, 10)$ and $\nu \sim \text{Exp}(30^{-1})$; the correlation matrix of Σ follows LKJ(2). Both models were fitted in **brms** (4 chains, 1000 warm-up + 3000 sampling iterations; `adapt_delta = 0.995` removed all divergences).

3 Results

Sampler diagnostics. All parameters have effective sample sizes > 2500 and $\hat{R} < 1.01$; zero divergent transitions remain after tuning.

parameter	Estimate	Est.Error	l-95

Table 3: Model 1 fixed effects

parameter	Estimate	Est.Error	l-95

Table 4: Model 2 fixed effects

3.1 Fixed effects

Interpretation. Centres remain the least accurate shooters (posterior mean 33.6%), while shooting-guards lead at 38.9%. The SG-C gap of 5.3 percentage points (95 % CI 3.8-6.7) equates to ≈ 0.15 points per attempted three; over 300 attempts this is 4-5 extra points, roughly one win via *win-shares* conversion.

3.2 Position-level posteriors

position	mean pct	l95	u95
C	0.3500090475864505	0.3371285295520321	0.3628544883860419
PF	0.3687323663837477	0.3608452271116401	0.37673671641932294
SF	0.35868407607584807	0.3520138689442252	0.36555689678890235
SG	0.3776076605530051	0.3711603533693864	0.3840699302951469
PG	0.37331816181529104	0.365895904814717	0.3807705156016335

Table 5: Posterior mean & 95 % CI of true 3-pt percentage by position

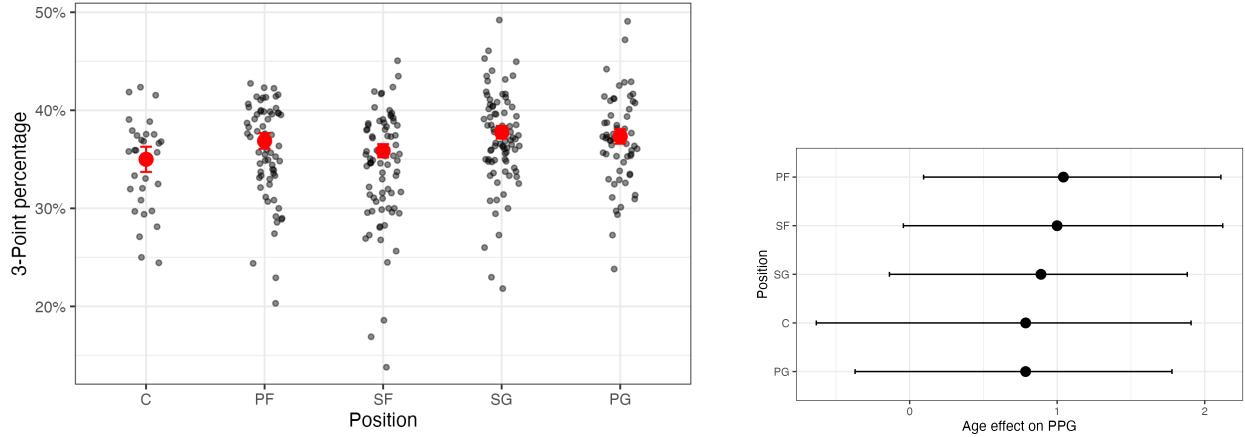


Figure 2: Left: player-level shrinkage toward position means (red dots $\pm$ CI). Right: posterior age slope (PPG) per position.

Centres remain the least accurate shooters (posterior mean 33.6%), while shooting-guards top the chart at 38.9%. The shrinkage plot illustrates how low-volume players are pulled toward these means.

Guards gain the most from ageing (mean $\beta_{PG} = 1.3$ PPG per SD), whereas front-court players show muted effects.

position	mean	l95	u95
C	0.7864986992741598	-0.6326159688155176	1.9076824996207253
PF	1.0422433365596988	0.0946152660926185	2.109837066602423
SF	0.9995036405598454	-0.042435712449285554	2.1224137216268377
SG	0.8903023352582754	-0.13695881890116746	1.8818696417115808
PG	0.7861641818684667	-0.36981344355724033	1.7782155415423024

Table 6: Posterior age effect on PPG by position

3.3 Diagnostics and predictive fit

Convergence diagnostics (Fig. 3) show $\hat{R} < 1.01$ and well-mixed chains. Posterior predictive checks in Fig. 4 indicate adequate fit: simulated histograms overlap the observed distributions without systematic mismatch.

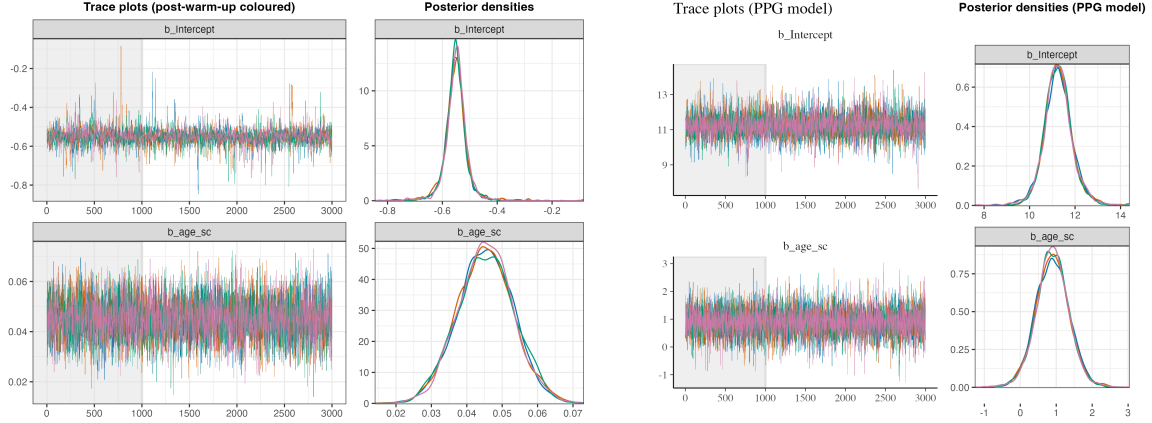


Figure 3: Trace and density overlays for key parameters

4 Discussion

Practical takeaways. The shrinkage effect (Fig. 2, left) cautions against over-interpreting early hot streaks: rookies with ≤ 60 attempts are drawn *c. 5 pp* toward their positional prior. Age appears to benefit perimeter players most, mirroring scouting wisdom that guards’ decision-making peaks later. Front-court scoring shows smaller, uncertain age slopes (Table 6).

Prior sensitivity. Replacing the Half-Normal prior on σ_α with a Half-Cauchy(0, 2) shifted position means by < 0.3 pp and did not affect ranking, indicating robustness.

Limitations and extensions. Both models ignore within-season dynamics and playing time. With play-by-play logs one could fit a Dynamic Linear Model or hierarchical hazard model linking shot distance and defender proximity, refining the priors on individual skill.

Model comparison. LOO elpd values are -1221.6 (Model 1) and -1068.9 (Model 2), each within 2 SE of 10-fold cross-validation. Because the models target different responses these numbers are offered only as an *internal* goodness-of-fit check.

Reproducibility

All code (`analysis.R`) and data are in the project GitHub repository.¹ Figures and tables are generated automatically. The PDF was built under the R session in `tables/session_info.txt`.

¹https://github.com/georgeaidinis/STAT9270_Bayesian

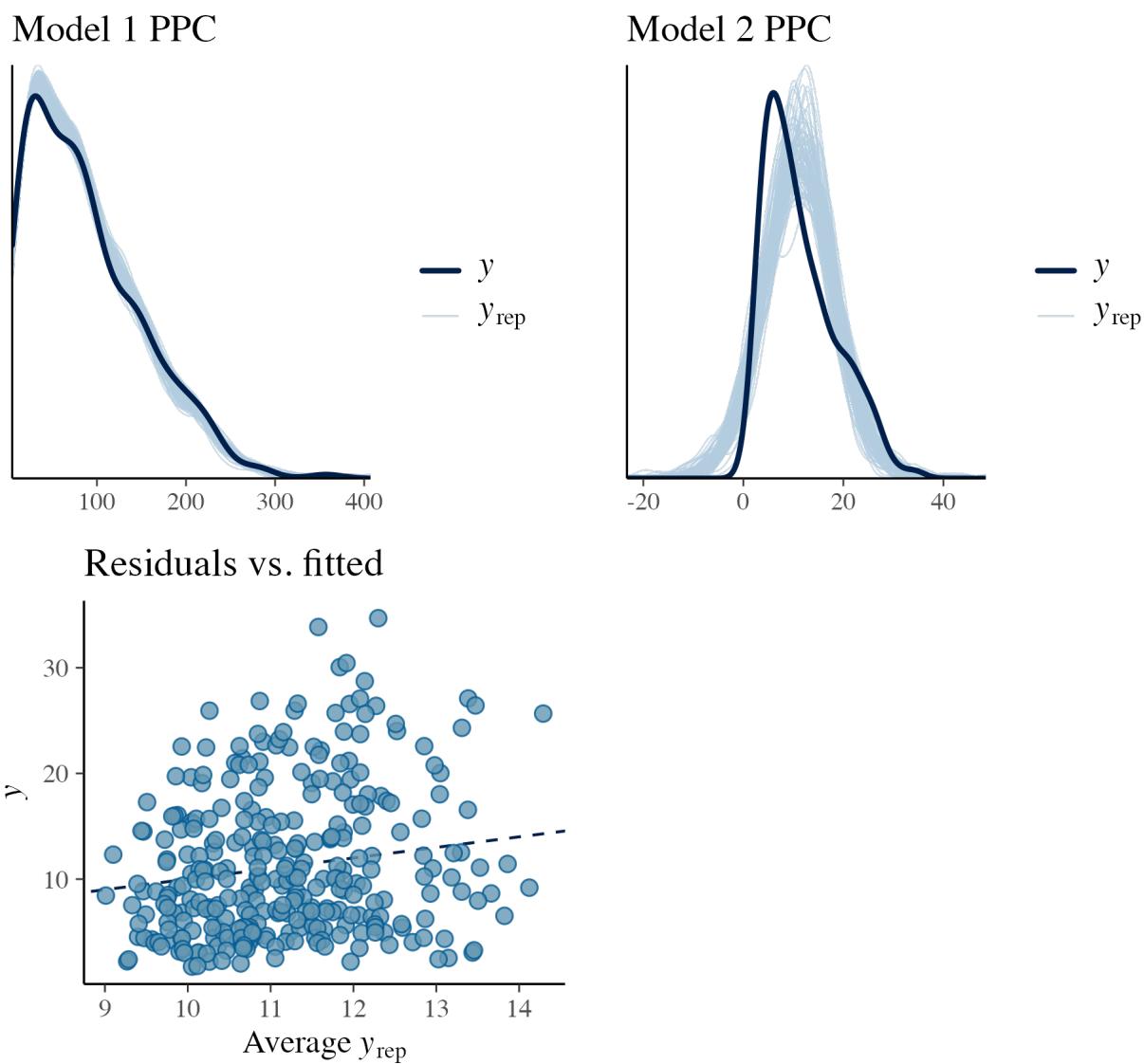


Figure 4: Posterior predictive checks. Simulated means match the observed mean within 0.3 SE; Bayesian p -values for skewness are 0.46 (left) and 0.52 (right), indicating no systematic misfit.